King Fahd University of Petroleum & Minerals
College of Computing and Mathematics
Information and Computer Science Department

ICS 253
Discrete Structures
Fall 2026 (261)

Instructors

Name	Contact	Venue
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Course Description:

Propositional Logic, Propositional Equivalence, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference; Methods of Proof, Divisibility and the Fundamental Theorem of Arithmetic; Sets, Set Operations, Cardinality of Sets; Functions; Recurrence Relations, Solving Recurrence Relations, Equivalence Relations and Congruences, Sequences and Summations; Mathematical Induction, Strong Induction, Recursive Definitions and Structural Induction, Well-Ordering Principle; Basics of Counting, Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients

Course Pre-requisites:

ICS 104: Intro. to Prog. in Python & C

Course Objectives:

1. To develop mathematical and thinking skills necessary for reading, comprehending, and constructing mathematical arguments.
2. To learn the fundamental concepts and techniques of discrete mathematics needed for problem solving in computer science.

Course Learning Outcomes:

1. Formulate and derive propositional/predicate logic expressions.
2. Apply proving methods.
3. Use theoretical principles of sets and functions to describe a computing task.
4. Apply counting techniques to solve combinatorial problems.

5. Formulate and solve recurrence relations.

Learning Resources:

- ICS 253 Course on https://www.zybooks.com/ (will be shared with students through blackboard)
- Discrete Mathematics and Its Applications. Rosen, Kenneth H. 7th Edition.

Course Contents:

The following schedule is tentative and subjected to changes. Any change will be announced in the course Blackboard. The chapter and section numbers are based on the **zybook**

Week#	Topics	
1, 2, 3, 4	**1. Logic** 1.1 Propositions and logical operations 1.2 Evaluating compound propositions 1.3 Conditional statements 1.4 Logical equivalence 1.5 Laws of propositional logic 1.6 Predicates and quantifiers 1.7 Quantified statements 1.8 De Morgan's law for quantified statements 1.9 Nested quantifiers 1.10 More nested quantified statements 1.11 Logical reasoning	1.12 Rules of inference with propositions 1.13 Rules of inference with quantifiers **2. Proofs** 2.1 Mathematical definitions 2.2 Introduction to proofs 2.3 Best practices and common errors in proofs 2.4 Writing direct proofs 2.5 Proof by contrapositive 2.6 Proof by contradiction 2.7 Proof by cases
5, 6, 7	**3. Sets** 3.1 Sets and Subsets 3.2 Set of sets 3.3 Union and intersection 3.4 More set operations 3.5 Set identities 3.6 Cartesian products	**4. Functions** 4.1 Definition of functions 4.2 Floor and ceiling functions 4.3 Properties of functions 4.4 The inverse of a function 4.5 Composition of Functions 4.6 Cardinality of Sets
8, 9	**5. Number Theory** 5.1The Division Algorithm 5.2 Greatest common divisor and Euclid's algorithm	5.3 Modular arithmetic 5.4 Fast exponentiation
10, 11	**6. Induction and Recursion** 6.1 Mathematical induction 6.2 More inductive proofs	6.3 Strong induction and well-ordering 6.4 Recursive definitions
12, 13	**7. Counting** 7.1 Sum and product rules 7.2 The bijection rule 7.3 The generalized product rule 7.4 Inclusion-exclusion principle 7.5 The pigeonhole principle 7.6 Counting permutations	7.7 Counting subsets 7.8 Subset and permutation examples 7.9 Counting by complement 7.10 Permutations with repetitions 7.11 Assignment problems: Balls in bins 7.12 Binomial coefficients and combinatorial identities

14, 15	**8. Relations** 8.1 Summations 8.2 Sequences 8.3 Recurrence relations 8.4 Solving linear homogeneous recurrence relations

Evaluation

Attendance	5%
Assignments (Submitted on Gradescope)	20%
Quizzes (in-class)	15%
Midterm	25%
Final exam (comprehensive)	35%

Course policies

Coursework includes participation, online/in-class discussions and activities, attendance, quizzes, and projects. Students are expected to be positively engaged in the learning process

Course Blackboard, MS Teams, and Gradescope: Students are required to periodically check the course Blackboard and download course material as needed.

- Several resources will be posted through the Blackboard.
- Blackboard will be used for calendar communication, posting grades, etc.
- You may use MS Teams to talk to your classmates and your instructor.
- Gradescope will be used to submit your homework assignments.

Class Attendance
Students are expected to arrive on time for class, consistent with expectations in professional settings. Recording attendance for another student, signing another student's attendance, or asking another student to record yours will be reported for investigation and may result in severe consequences for both students involved.

- Attendance will be checked at each lecture.
- Missing 20% of the classes will result in an automatic DN grade (without warning).
- Late arrivals will disrupt the class session and may be counted as a miss if repeated.
- Every unexcused absence may lead to a loss of 0.5% of total grade.

Late assignments are rejected or subjected to late penalty. See late submission policy on the Blackboard/Gradescope.

Re-grading policy: If you have a complaint about any of your grades, discuss it with the instructor first, and officially submit a regrade request on Gradescope no later than 3 days of distributing the grades (except for the final). Only legitimate concerns on grading should be discussed.

Office Hours: Students are encouraged to use the office hours to clarify any part of the material that is not clear.

Courtesy:

- Students are expected to be courteous toward their classmates and the instructor throughout the duration of this course (in-class and online).
- Side-talks and text-messages during the class are prohibited.

Honor code

Students of this course are expected to maintain the highest level of academic integrity and ethics. A KFUPM student does not cheat or tolerate those who do. Every student is expected to submit work that is entirely their own, to properly cite all sources used, and to refrain from any form of academic misconducts.

Plagiarism

Use of another person's ideas, processes, results, or words requires appropriate credit or reference to the author. Use of Generative AI tools is subject to the instructor's authorization. Submitting the same work in more than one course without the instructor's permission is not allowed.

Examinations Policies

You are expected to adhere to KFUPM policies for all examinations in this course, including quizzes, major exams, midterms, and final exams. The following policies are enacted in this course during examinations:

- Students are expected to arrive at the examination hall on time with proper University ID.
- Students are seated according to the instructor's seat assignment and may request a seat change only for legitimate reasons.
- Late arrival to the exam may result in the student being denied admission. Once any student has begun finishing the exam, no further admission will be allowed under any circumstance.
- Students are not permitted to leave the examination room to use the restroom during an exam, except in rare cases involving a documented medical condition (e.g., diabetes, kidney disease). Any excused restroom leave will be fully supervised by a same-gender proctor.
- Students must leave all smart devices outside the examination hall, including smartphones, smartwatches, earbuds, and unauthorized calculators. Possession of any unauthorized device will be considered a cheating attempt and will be handled accordingly.
- Cheating attempts will be reported according to KFUPM policy and will result in a DN grade in the course, along with potential further penalties upon investigation.